CASE STUDY: BRAYS BAYOU FLOOD FREQUENCY

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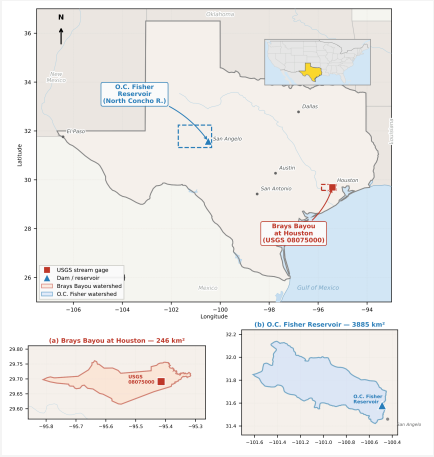
TODAY

We discuss Smith et al. (2026), a nonstationary flood frequency analysis of Brays Bayou at Houston, fit by Bayesian methods in the open-source RMC-BestFit software from the US Army Corps of Engineers.

! Important

Every figure is from the paper, which is open access under CC BY 4.0, so it's OK to share here.

CASE STUDY: TWO TEXAS WATERSHEDS



Brays Bayou and O.C. Fisher Reservoir (Smith et al., 2026, Fig. 1, CC BY 4.0).

Clear physical drivers and strong nonstationary signals

- **Brays Bayou:** floods rising with urbanization
- **O.C. Fisher:** floods falling with brush encroachment and groundwater pumping

The data

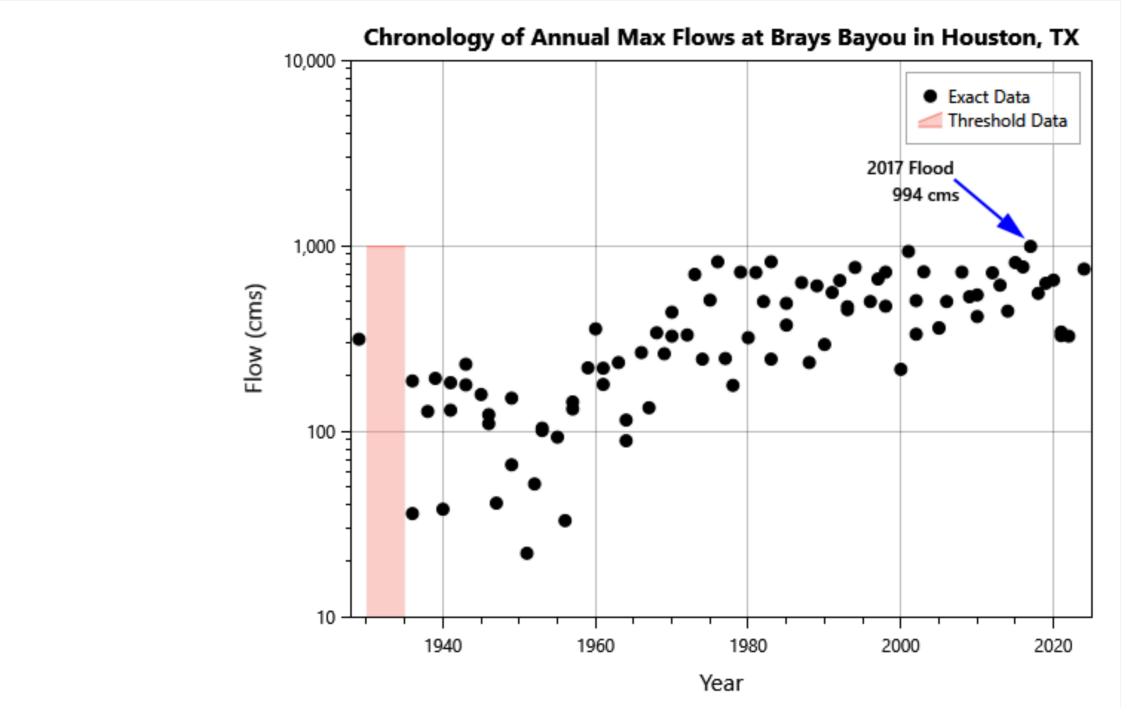
The model

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THE RECORD IS A MODELING CHOICE



Annual maximum instantaneous flow at USGS 08075000, 1929-2024; the shaded band is 1930-1935, when only a perception threshold is known (Smith et al., 2026, Fig. 2, CC BY 4.0).

THE LIKELIHOOD

The stationary likelihood (Smith et al., 2026, eq. 1):

$$L(\mathbf{x} \mid \theta) = \prod_{i=1}^{N} f(x_i \mid \theta)$$

In the **nonstationary model** the parameters depend on the year, θ_t (Smith et al., 2026, eq. 2):

$$L(\mathbf{x} \mid \theta) = \prod_{t=1}^{T} f(x_t \mid \theta_t)$$

A PERCEPTION THRESHOLD IS A CENSORED OBSERVATION

A perception threshold is “the flow above which an event would have been recorded in any given year” (Smith et al., 2026, p. 11, eq. 3).

$$L(\mathbf{x} \mid \theta) = \prod_{t=1}^{T} \begin{cases} f(x_t \mid \theta_t) & \text{exact data} \\ F(x_t^U \mid \theta_t) & \text{left censored (below threshold)} \\ 1 - F(x_t^L \mid \theta_t) & \text{right censored (above threshold)} \\ F(x_t^U \mid \theta_t) - F(x_t^L \mid \theta_t) & \text{interval censored} \end{cases}$$

f is the probability density and F the cumulative distribution function.

BRAYS BAYOU LIKELIHOOD

- 1929 and 1936-2024: $f(x_t \mid \theta_t)$.
- Each of 1930-1935: $F(994 \mid \theta_t)$.

A year with no measured flood still carries information: six years in a row below 994 m³/s rules out parameter values that make 994 m³/s common.

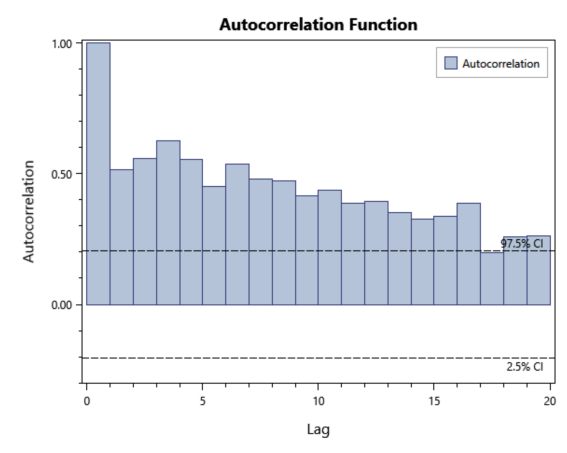
For more, read Gelman et al. (2014), §8.7, on censored data in a Bayesian model.

NHST

Subperiod tests compare 1929-1980 against 1981-2024 (Smith et al., 2026, p. 13, Table 2).

Test	Null hypothesis	p-value
Jarque-Bera	Normality	0.0015
Ljung-Box Q	No autocorrelation	$< 10^{-15}$
Wald-Wolfowitz	Independence and stationarity	1.84×10^{-10}
Mann-Whitney U	Homogeneity	4.55×10^{-11}
Mann-Kendall	No monotonic trend	1.33×10^{-13}
Linear trend	$\beta_1 = 0$	$< 10^{-15}$
Equal-variance t-test	$\mu_1 = \mu_2$	4.23×10^{-13}
Unequal-variance t-test	$\mu_1 = \mu_2$	5.91×10^{-12}
F-test	$\sigma_1^2 = \sigma_2^2$	8.76×10^{-7}
Mixture model	Unimodality	0.0004

THE AUTOCORRELATION OF A TRENDING RECORD



Autocorrelation of the Brays annual maxima out to lag 20, with 95% bounds (Smith et al., 2026, Fig. 3, CC BY 4.0).

- Nearly every lag out to 20 sits above the bound.
- A trend does this: early years are all low and late years all high, so any two years close in time look alike.
- The paper reads it as “characteristic of a strong monotonic trend” (Smith et al., 2026, p. 13).

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LOG-PEARSON TYPE III

LP3 is used “consistent with federal guidelines established in Bulletin 17C” (Smith et al., 2026, p. 8).

- Like the GEV, it is fit to annual maxima.
- “Log” means it is fit to $\log_{10}$ of the flow.
- Think: log-normal with a skewness parameter added.
  - Positive means prone to floods far above the mean
  - Negative means the watershed caps the largest flow.
  - The skew is often similar at nearby gauges – we will revisit this next week!

TREND FORMULAS THE SOFTWARE OFFERS

Model	Formulation	Use case in the paper
Constant	$\mu_t = \alpha$	No detectable trend
Linear	$\mu_t = \alpha + \beta t$	Gradual urban expansion
Exponential	$\mu_t = \alpha e^{-\beta t}$	Trend slowing as a watershed stabilizes
Logistic	$\mu_t = \frac{\alpha}{1+e^{-\beta t}}$	Sprawl that plateaus
Power	$\mu_t = \alpha t^{\beta}$	Accelerating imperviousness
Quadratic	$\mu_t = \alpha + \beta t + \gamma t^2$	Rise then plateau
Cubic	$\mu_t = \alpha + \beta t + \gamma t^2 + \delta t^3$	Sequential land-use changes
Sinusoidal	$\mu_t = \alpha + \beta \sin(2\pi\gamma t + \delta)$	Multidecadal climate cycles
Step	$\mu_t = \mu_1$ for $t \leq t_c$, μ_2 for $t > t_c$	A land-cover change in a known year

TIME IS THE COVARIATE

Think

Monday's recipe said to choose a covariate by mechanism. What does t *stand in* for at Brays?

- Imperviousness and channelization, neither of which is in the model.
- The authors say so: "formal covariate-based attribution is beyond the current scope" (Smith et al., 2026, p. 13).

BAYES' THEOREM IN SMITH'S NOTATION

Smith et al. (2026), eq. 4:

$$p(\theta \mid \mathbf{x}) = \frac{L(\mathbf{x} \mid \theta) \cdot \pi(\theta)}{\int L(\mathbf{x} \mid \theta) \cdot \pi(\theta) d\theta}$$

The joint prior over p parameters, then with the parameters independent (Smith et al., 2026, eqs. 5-6):

$$\pi(\theta) = \pi(\theta_1 \cap \theta_2 \cap \cdots \cap \theta_p)$$

$$\pi(\theta) = \prod_{i=1}^{p} f(\theta_i)$$

THE DEFAULT PRIORS

- Flat priors on location, scale, and shape, “centered near the likelihood with broad spread” (Smith et al., 2026, p. 5).
- A Jeffreys prior on the scale, $\pi(\sigma) \propto 1/\sigma$, “equivalent to using a log-link function” (p. 5).
- That is a flat prior on $\log \sigma$: the same log scale Monday’s model used for σ.
- Trend parameters (slopes, change points) get uninformative priors, and priors on the other parameters apply at the final year T (p. 6).

PRIOR INFORMATION ON A FLOOD QUANTILE

A quantile prior puts a distribution on the flood x_p with annual exceedance probability p, with hyperparameters θ_h (Smith et al., 2026, eq. 7):

$$h(x_p) = f(x_p \mid \theta_h)$$

The quantile is a function of the parameters in the final year (Smith et al., 2026, eq. 8):

$$x_p = F^{-1}(p \mid \theta_T)$$

A QUANTILE PRIOR BECOMES A PRIOR ON THE PARAMETERS

Substitute to evaluate it in parameter space (Smith et al., 2026, eq. 9):

$$\pi_q(\theta) = h\left(F^{-1}(p \mid \theta_T)\right)$$

The posterior multiplies in every source (Smith et al., 2026, eq. 10):

$$p(\theta \mid \mathbf{x}) \propto L(\mathbf{x} \mid \theta) \cdot \pi(\theta) \cdot \pi_q(\theta)$$

“The quantile prior acts as a penalty function, rewarding parameter sets that produce flood quantiles consistent with the prior information” (p. 6). O.C. Fisher uses one, taken from a rainfall-runoff model; Brays does not.

SAMPLING THE POSTERIOR

RMC-BestFit samples with DE-MCzs, a differential-evolution Markov chain with a “snooker update” (Smith et al., 2026, p. 9).

- It uses no gradients.
- It scales poorly to many parameters, where the NUTS sampler you used in `Turing` scales well (by leveraging gradients)
- It can do better on a posterior with several modes, or a likelihood with discrete parameters, such as the step model’s change year t_c.

THE POSTERIOR PREDICTIVE DISTRIBUTION

The full distribution of a flood k years into the future (Smith et al., 2026, eq. 14):

$$p(x_{T+k} \mid \mathbf{x}) = \int f(x_{T+k} \mid \theta_{T+k}) \cdot p(\theta \mid \mathbf{x}) d\theta$$

- It "accounts for both parameter uncertainty (epistemic) and natural variability (aleatory)" (p. 10), as in labs 4 and 5.
  - This is a *narrow* definition of epistemic uncertainty
- Think of it as "averaging over the posterior"

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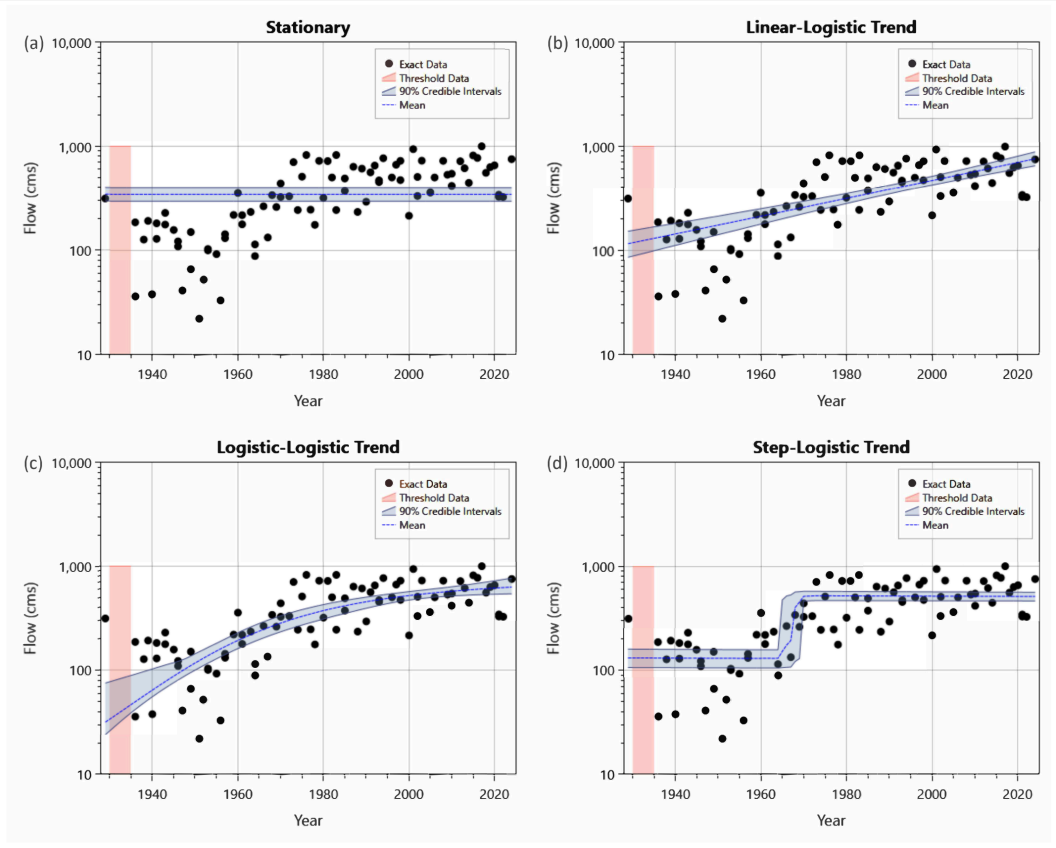
Discussion

SEVEN CANDIDATE MODELS FOR BRAYS

Model comparison (Smith et al., 2026, p. 15, Table 3), with AIC, BIC, and DIC revisited in week 8.

Model	k	AIC	BIC	DIC	ΔDIC
Stationary	3	1233.73	1241.23	1228.36	73.06
Linear (μ)	4	1182.14	1192.14	1180.76	25.45
Logistic (μ)	4	1179.94	1189.94	1179.23	23.93
Step (μ)	5	1177.04	1189.54	1162.28	6.98
Linear-logistic (μ-σ)	5	1168.70	1181.20	1172.04	16.74
Logistic-logistic (μ-σ)	5	**1161.46**	**1173.96**	1160.26	4.95
Step-logistic (μ-σ)	6	1165.24	1180.24	**1155.30**	0.00

FOUR PLAUSIBLE MODELS



The 2-year return level under (a) the stationary, (b) linear-logistic, (c) logistic-logistic, and (d) step-logistic models, with 90% credible intervals (Smith et al., 2026, Fig. 4, CC BY 4.0).

THE FITTED CHANGE POINT AND THE VARIANCE

- The change point is estimated at 1968, near the start of Houston's most intensive development in the early 1970s (Smith et al., 2026, p. 14).
- The location jumps at the change point, while the scale, the spread of the log-flows, declines smoothly over the whole record with no jump (logistic, $\beta = -0.014$).
- Early floods spanned about 20-300 m^3/s, more than a factor of ten; after urbanization, about 200-800 m^3/s, a factor of four (p. 14).

WHY URBANIZATION NARROWS THE LOG-SPACE SPREAD

“In a pre-urbanization watershed, flood magnitudes are highly variable because antecedent soil moisture, infiltration capacity, and vegetation interception modulate rainfall-runoff partitioning differently for each event” (Smith et al., 2026, p. 23).

Pave it over and the response “approaches a near-deterministic conversion of rainfall to runoff” (p. 24).

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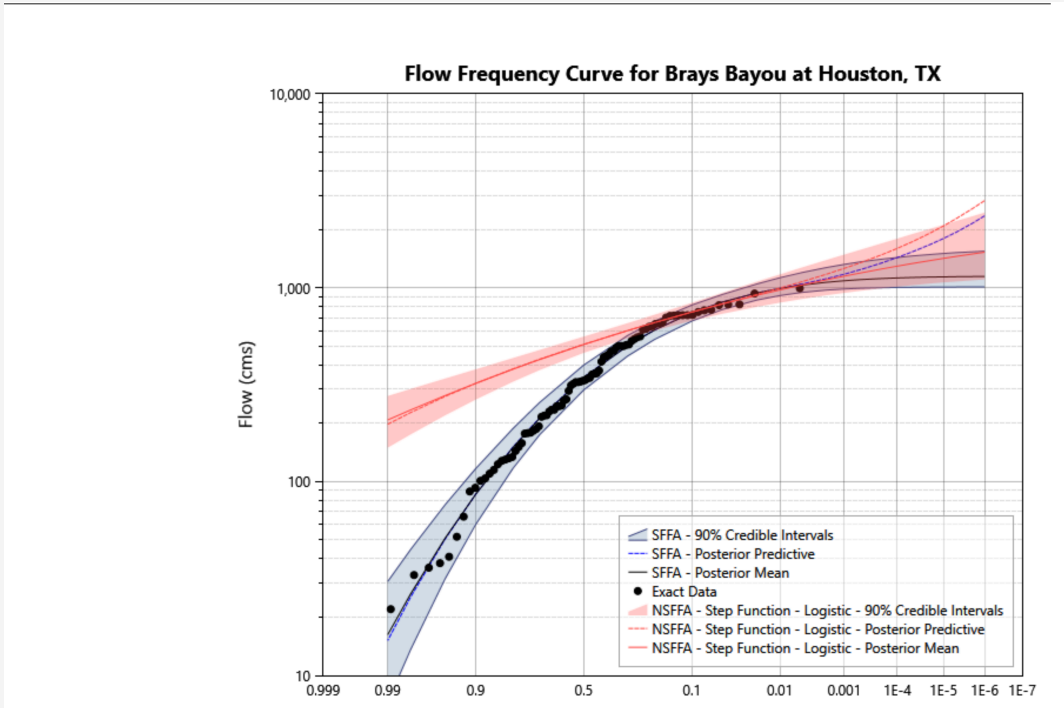
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STATIONARY AND NONSTATIONARY FLOOD QUANTILES

Posterior predictive estimates and 90% intervals in m^3/s, the nonstationary column at 2024 conditions (Smith et al., 2026, p. 16, Table 4).

AEP (return period)	Stationary	90% CI	Nonstationary	90% CI
0.50 (2-yr)	346	[297, 400]	511	[462, 560]
0.20 (5-yr)	607	[542, 673]	661	[603, 727]
0.10 (10-yr)	744	[674, 815]	749	[675, 836]
0.02 (50-yr)	940	[865, 1046]	924	[798, 1076]
0.01 (100-yr)	993	[912, 1126]	996	[838, 1172]

FREQUENCY CURVES MEET IN THE TAIL



Stationary (blue) and nonstationary (red) frequency curves for Brays Bayou, with 90% intervals; the plotting positions assume stationarity (Smith et al., 2026, Fig. 5, CC BY 4.0).

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STEP MODEL

Think

The selected model has one level before 1968 and one after, forever. What would violate that?

- More urbanization after 1968.
- More infrastructure: channelization and detention basins.

HOW YOU MIGHT ADDRESS IT

- A better covariate: impervious area fraction in place of the year, which the paper names as future work (Smith et al., 2026, p. 25).
- A different method: fit the extreme value model to rainfall, and run it through a flood model that represents land use and infrastructure.
- The paper's own list adds hierarchical pooling across sites, model averaging over the trend forms, and alternative distributions (pp. 24-25).